

Jonathan Montaña

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INTERESTS

Commutative algebra, combinatorics, and algebraic geometry.

EXPERIENCE

Associate Professor	Arizona State University	August 2025-
Assistant Professor	Arizona State University	August 2022-July 2025
Research Member	SLMath(MSRI), Berkeley, CA	January 2024-April 2024
Assistant Professor*	New Mexico State University	August 2018-July 2022
*Granted tenure and promoted to Associate Professor in the spring of 2022		
Oberwolfach Leibniz Fellow (postdoc)	MFO, Oberwolfach Res. Inst. for Math.	Summer 2018
Visiting Assistant Professor (postdoc)	University of Kansas	August 2015-May 2018
Graduate Assistant	Purdue University	August 2009-July 2015
Program Associate (student member)	MSRI, Berkeley, CA	August 2012-December 2012
Graduate Assistant	Universidad Nacional de Colombia, Bogotá	August 2008-August 2009

EDUCATION

Purdue University, West Lafayette, IN
Ph.D. in Mathematics, August 2015 (Advisor: Bernd Ulrich)
M.Sc. in Mathematics, August 2011

Universidad Nacional de Colombia, Bogotá
M.Sc. in Mathematics, August 2009
B.Sc. in Mathematics, August 2008

PUBLICATIONS

Mathematics:

31. *The multiplicity sequence of monomial ideals*, with Sudipta Das and Aniketh Sivakumar, **arXiv: 2608.12731, submitted.**
30. *Matroid correspondence*, with Colin Crowley, Changxin Ding, Haoxi Hu, Arvind Kumar, Thai Thanh Nguyen, Anna Shapiro, Noah Solomon, Botong Wang, and Juliet Whidden, **arXiv: 2607.13783, submitted.**
29. *Homology of Vietoris-Rips complexes of hypercube graphs via group actions*, with Federico Galetto and Zoe Wellner, **arXiv: 2606.20784, submitted.**
28. *Quasipolynomial behavior via constructibility in multigraded algebra*, with Hailong Dao, Ezra Miller, Christopher O'Neill, and Kevin Woods, **arXiv: 2512.18536, submitted.**
27. *Rational powers, invariant ideals, and the summation formula*, with Sankhaneel Bisui, Sudipta Das, and Tài Huy Hà, **arXiv: 2402.12350, submitted.**
26. *K-polynomials of multiplicity-free varieties*, with Federico Castillo, Yairon Cid-Ruiz, and Fatemeh Mohammadi, **arXiv: 2212.13091, submitted.**
25. *Degree functions of graded families of ideals*, with Steven Dale Cutkosky, **Collectanea Mathematica**, to appear, **arXiv: 2506.05248.**

24. *Multiplicities and degree functions in local rings via intersection products*, with Steven Dale Cutkosky, **Journal of the London Mathematical Society**, 112 (2025), e70338.
23. *Ladder determinantal varieties and their symbolic blowups*, with Alessandro De Stefani, Luis Núñez-Betancourt, Lisa Seccia, and Matteo Varbaro, **Mathematische Zeitschrift** 312:92 (2026).
22. *The core of monomial ideals*, with Louiza Fouli, Claudia Polini, and Bernd Ulrich, **Algebra & Number Theory**, 19 (2025), 1463-1494.
21. *Blowup algebras of determinantal ideals in prime characteristic*, with Alessandro De Stefani and Luis Núñez-Betancourt, **Journal of the London Mathematical Society**, 110 (2024), e12969.
20. *Computing multiplicity sequences*, with Justin Chen and Youngsu Kim, **Journal of Software for Algebra and Geometry** 14 (2024), 13-18.
21. *Double Schubert polynomials do have saturated Newton polytopes*, with Federico Castillo, Yairon Cid-Ruiz, and Fatemeh Mohammadi, **Forum of Mathematics, Sigma** 11 (2023), e100.
18. *Purity of monoids and characteristic-free splittings in semigroup rings*, with Alessandro De Stefani and Luis Núñez-Betancourt, **Mathematische Zeitschrift** 305:31 (2023).
17. *Frobenius methods in combinatorics*, with Alessandro De Stefani and Luis Núñez-Betancourt, **São Paulo Journal of Mathematical Sciences** 17 (2023), 387-429.
16. *Convex bodies and graded families of monomial ideals*, with Yairon Cid-Ruiz, **Revista Matemática Iberoamericana** 38 (2022), 2033-2056.
15. *Mixed multiplicities of graded families of ideals*, with Yairon Cid-Ruiz, **Journal of Algebra** 590 (2022), 394-412.
14. *When are multidegrees positive?*, with Federico Castillo, Yairon Cid-Ruiz, Binglin Li, and Naizhen Zhang, **Advances in Mathematics** 374 (2020), 107382.
13. *Exterior powers and Tor-persistence*, with Justin Lyle and Keri Sather-Wagstaff, **Journal of Pure and Applied Algebra** 226 (2022), 106890.
12. *Analytic spread and integral closure of integrally decomposable modules*, with Carles Bivià-Ausina, **Nagoya Mathematical Journal** 245 (2022), 166-191.
11. *Symbolic analytic spread: upper bounds and applications*, with Hailong Dao, **Journal of the Institute of Mathematics of Jussieu** 20 (2021), 1969-1981.
10. *Extremal growth of Betti numbers and trivial vanishing of (co)homology*, with Justin Lyle, **Transactions of the American Mathematical Society** 373 (2020), 7937-7958.
9. *Splittings and symbolic powers of square-free monomial Ideals*, with Luis Núñez-Betancourt, **International Mathematics Research Notices** 2021 (2021), 2304-2320.
8. *On asymptotic vanishing behavior of local cohomology*, with Hailong Dao, **Mathematische Zeitschrift** 295 (2020), 73-86.
7. *Length of local cohomology of powers of ideals*, with Hailong Dao, **Transactions of the American Mathematical Society** 371 (2019), 3483-3503.
6. *Lower bounds on projective levels of complexes*, with Hannah Altmann, Eloísa Grifo, William Sanders, and Thanh Vu, **Journal of Algebra** 491 (2017), 343-356.
5. *On the growth of deviations*, with Adam Boocher, Alessio D'Alì, Eloísa Grifo, and Alessio Sammartano, **Proceedings of the American Mathematical Society** 144 (2016), 5049-5060.
4. *Edge ideals and DG algebra resolutions*, with Adam Boocher, Alessio D'Alì, Eloísa Grifo, and Alessio Sammartano, **Le Matematiche** 70 (2015), no.1, 215-237.
3. *Artin-Nagata properties, minimal multiplicities, and depth of fiber cones*, **Journal of Algebra** 425 (2015), 423-449.
2. *Multiplicities of classical varieties*, with Jack Jeffries and Matteo Varbaro, **Proceedings of the London Mathematical Society** 110 (2015), 1033-1055.
1. *The j -multiplicity of monomial ideals*, with Jack Jeffries, **Mathematical Research Letters** 20 (2013), 729-744.

Computer Science:

4. *Safe Explicable Policy Search*, with Akkamahadevi Hanni and Yu Zhang, **IEEE International Conference on Robotics and Automation (ICRA) 2026**.
3. *Improving Locality Sensitive Hashing by Efficiently Finding Projected Nearest Neighbors*, with Omid Jafari and Parth Nagarkar, **13th International Conference on Similarity Search and Applications (SISAP) 2020**.
2. *mmLSH: A Practical and Efficient Technique for Processing Approximate Nearest Neighbor Queries on Multimedia Data*, with Omid Jafari and Parth Nagarkar, **13th International Conference on Similarity Search and Applications (SISAP) 2020**.
1. *Efficient Bitmap-based Indexing and Retrieval of Similarity Search Image Queries*, with Omid Jafari and Parth Nagarkar, **IEEE Southwest Symposium on Image Analysis and Interpretation (SSIAI) 2020**.

CODES

MultiplicitySequence package in Macaulay2, with Justin Chen and Youngsu Kim, **certified by JSAG**.

GRANTS

NSF Standard Grant, DMS #2401522	NSF, \$225,000	2024-2027
NSF Standard Grant, DMS #2001645/2303605	NSF, \$139,794	2020-2024
NSF Conference Grant, DMS #2502634	SLAM 2025, \$20,000	March 2025
NSA Conference Grant	SLAM 2024, \$20,368	February 2024
NSF Conference Grant, DMS #2302498	SLAM 2023, \$16,000	March 2023
Natural Sciences DEI Seed Grant	ASU, \$10,000	2023
NSF Conference Grant, DMS #2220824	PASCA 2022, \$11,900	June 2022
AMS - Simons Travel Grant	AMS & Simons Foundation, \$4,000	July 2017-June 2019

FELLOWSHIPS, HONORS, AND ACADEMIC VISITS

Member	AxIOM program, SLMath(MSRI)	February 2027
Charles Wexler Teaching Award	SoMSS, ASU	April 2026
Research Member	SLMath(MSRI)	January-April 2024
AIM SQuaREs	AIM	2023-2025
Oberwolfach Research Fellow	MFO, Oberwolfach Res. Inst. for Math.	November-December 2022
Oberwolfach Leibniz Fellow	MFO, Oberwolfach Res. Inst. for Math.	Summer 2018
Bilsland Dissertation Fellowship	Purdue University, College of Science	June 2014-May 2015
Program Associate	MSRI	August-December 2012
Ross Fellowship	Purdue University	August 2009-July 2010
Outstanding Graduate Students Fellowship	Universidad Nacional de Colombia, Bogotá	August 2008- July 2009
Second place (nationwide)	ECAES in Mathematics, Colombia	2008

INVITED TALKS

Singularity Theory Day	Univ. Politècnica de València, Spain	July 7, 2026
The ubiquity of commutative algebra	CIRM, France	March 19, 2026
Macaulay2 Winter School (lecture series)	Hue City, Vietnam	Dec. 29 - Jan. 2, 2026
AMS Sectional Meeting (2 talks)	Tulane University	October 4, 2025
Algebra and Combinatorics Seminar	Tulane University	October 1, 2025
Mathematical Congress of the Americas	Miami, Florida	July 23, 2025
50th Spring Lecture Series	University of Arkansas	May 15, 2025
Alg., Comb., and NT Seminar	DePaul University	May 9, 2025
Alg., Comb., & Disc. Geom. Workshop	IPAM, UCLA	February 10, 2025
AMS Sectional Meeting	University of California, Riverside	October 26, 2024
Latin-American & Caribbean Math. Congress	João Pessoa, Brazil	August 26, 2024
Latin-American Alg. Colloquium (plenary)	Pont. Univ. Católica, Chile	July 22, 2024

CAAGTUS 2024 Conference	University of Arizona	May 4, 2024
Bay Area Discrete Math Day	Santa Clara University	April 20, 2024
Commutative Algebra Seminar	SLMath/MSRI	March 21, 2024
Algebra Seminar	Auburn University	October 16, 2023
ICMAM Latin-America, Satellite Conference	Online	September 27, 2023
Summer CAMP	University of Nebraska-Lincoln	August 9, 2023
MSRI summer school Colloquium	University of Notre Dame	May 26, 2023
Algebra Seminar	University of Missouri	April 17, 2023
Tulane Mathematics Colloquium	Tulane University	March 30, 2023
XII School of Mathematics (lecture series)	Fed. Univ. of Sergipe, Brazil (Online)	February 14-16, 2023
Algebra&Geometry seminar	University of Genova, Italy	December 21, 2022
Dept. of Mathematical Sciences Colloquium	University of Texas at El Paso (Online)	November 4, 2022
AMS Sectional Meeting	University of Utah	October 23, 2022
Comb. and Commalg. Seminar	Oklahoma State University (Online)	April 20, 2022
2022 JMM	Online	April 7, 2022
2022 JMM	Online	April 7, 2022
AMS Sectional Meeting	Online	March 26, 2022
AMS Sectional Meeting	Online	March 19, 2022
Southwest Local Alg. Mtg. (SLAM)	Baylor University	February 26, 2022
Sch. Math and Stat. Sc. Colloquium	Arizona State University	February 22, 2022
Dept. of Mathematics Millican Colloquium	University of North Texas	February 8, 2022
Dept. of Math., Stat., Comp. Sc. Colloquium	University of Illinois Chicago (Online)	December 7, 2021
AG-CA Seminar	University of Notre Dame (Online)	November 16, 2021
Commutative Algebra Seminar	U. of Illinois at Chicago (Online)	November 4, 2021
AMS Sectional Meeting	Online	October 23, 2021
Nat. Congress Mexican Math Soc.	Online	October 21, 2021
Algebra and Alg. Geometry Seminar	McMaster University, Canada (Online)	July 19, 2021
Virtual CA seminar	IIT Bombay, India (Online)	July 9, 2021
Fellowship of the Ring	MSRI (Online)	April 29, 2021
AMS Sectional Meeting	Online	April 17, 2021
Dept. of Math. Colloquium	New Mexico State University (Online)	April 9, 2021
Algebra and Geometry Seminar	Iowa State University (Online)	March 16, 2021
Dept. of Math. Colloquium	U. of California, Riverside (Online)	February 24, 2021
Combinatorics Seminar	University of Kansas (Online)	November 13, 2020
Dept. of Math. Colloquium	U. de los Andes, Colombia (Online)	October 22, 2020
Dept. of Math. Colloquium	University of New Mexico (Online)	October 13, 2020
AMS Sectional Meeting	Online	October 3, 2020
AG and CA seminar	NU/UIC/UofC (Online)	July 29, 2020
eCARs conference	Online	June 28, 2020
Algebra Seminar	U./Neu./Frib./Bern, Switzerland (Online)	May 15, 2020
AMS Sectional Meeting (meeting canceled)	Purdue University	April 4, 2020
Algebra and Algebraic Geometry Seminar	University of Wisconsin, Madison	February 7, 2020
Commutative Algebra Seminar	Purdue University	April 22, 2019
AMS Sectional Meeting	Auburn University	March 15, 2019
Frobenius Action in CA Workshop	Barcelona, Spain	January 18, 2019
Algebra Seminar	Universitat Pol. València, Spain	January 15, 2019
Fall School in CA	CIMAT, Mexico	November 13-17, 2018
Algebra Seminar	Georgia State University	September 11, 2018
Public Lecture at ECCO 2018	Barranquilla, Colombia	June 7, 2018
Col.-Mex. Math Soc. joint meeting	Universidad del Norte, Colombia	May 30, 2018
AMS Sectional Meeting	Portland State University	April 14, 2018
CA-AG Seminar	CIMAT, Mexico	March 22, 2018
Geometry and Topology Seminar	San Francisco State University	March 12, 2018
Commutative Algebra Seminar	University of Michigan	November 2, 2017
Commutative Algebra Seminar	University of Utah	October 19, 2017
PRIMA 2017	Oaxaca, Mexico	August 14, 2017

Combinatorial and CA Seminar	Oklahoma State University	April 20, 2017
Commutative Algebra Seminar	University of Nebraska-Lincoln	April 10, 2017
KUMUNU jr 2017	University of Nebraska-Lincoln	April 8, 2017
Algebra Seminar	University of Missouri	February 27, 2017
Department of Mathematics Colloquium	New Mexico State University	February 9, 2017
2016 CMS Winter Meeting	Niagara Falls, Canada	December 4, 2016
Algebra & Combinatorics Seminar	Tulane University	November 10, 2016
Commutative Algebra Seminar	Purdue University	October 21, 2016
KUMUNU 2016	University of Kansas	October 1, 2016
AMS Sectional Meeting	North Dakota State University	April 16, 2016
Algebra Seminar	University of Arkansas	January 29, 2016
AMS Sectional Meeting	Rutgers University	November 14, 2015
Combinatorics Seminar	University of Kansas	November 6, 2015
Commutative Algebra Seminar	Purdue University	October 7, 2015
Midwest Commutative Algebra Meeting	Purdue University	August 5, 2015
CA and related topics 2015	University of Kansas	March 22, 2015
Commutative Algebra Seminar	Purdue University	February 4, 2015
2015 Joint Mathematics Meeting	San Antonio, TX	January 13, 2015
Commutative Algebra Seminar	University of Utah	November 21, 2014
Commutative Algebra Seminar	Purdue University	November 5, 2014
Algebra Seminar	New Mexico State University	November 3, 2014
AMS Sectional Meeting	San Francisco State University	October 26, 2014
AMS Sectional Meeting	Dalhousie University	October 19, 2014
AMS Sectional Meeting	University of New Mexico	April 5, 2014
Commutative Algebra Seminar	Purdue University	February 27, 2014
Commutative Algebra Seminar	Purdue University	October 28, 2013
Commutative Ring Theory Seminar	University of Illinois at U-C	October 24, 2013
Commutative Algebra Seminar	Purdue University	October 21, 2013
AMS Sectional Meeting	University of Louisville	October 5, 2013
KUMUNU jr 2013	University of Nebraska-Lincoln	April 22, 2013
Commutative Ring Theory Seminar	University of Illinois at U-C	March 7, 2013
AG-CA Seminar	University of Notre Dame	February 26, 2013
Commutative Algebra Seminar	Purdue University	February 22, 2013

ORGANIZATIONAL ACTIVITIES

<i>Co-Organizer</i> , Number Theory&Algebra Seminar	Arizona State University	Fall 22-Spring 25, Fall 26 -
<i>Co-Organizer</i> , AMS Spe. Ses. on CA	Arizona State University	November 2026
<i>Co-Organizer</i> , Pan-American School on CA 2.0	CIMAT, Mexico	May 2026
<i>Co-Organizer</i> , AZ-NM Symp. on CA and Inter.	New Mexico State University	December 2025
<i>Co-Organizer</i> , Southwest Local Algebra Meeting	Yearly	2023-2025
<i>Co-Organizer</i> , Algebra Days at Arizona State	Arizona State University	November 2023
<i>Co-Organizer</i> , AMS Spe. Ses. on CA and Comb.	University of Texas, El Paso, TX	September 2022
<i>Co-Organizer</i> , Pan-American School on CA	CIMAT, Mexico	June 2022
<i>Co-Organizer</i> , AMS Spe. Ses. on CA	Online	September 2020
<i>Co-Organizer</i> , Algebra Seminar	New Mexico State University	Fall 2019-Fall 2021
<i>Co-Organizer</i> , CAAG Reading Seminar	New Mexico State University	Spring 2019-
<i>Co-Organizer</i> , AMS Spe. Ses. on Multiplicities	University of Michigan, Ann Arbor, MI	October 2018
<i>Co-Organizer</i> , Algebra Seminar	University of Kansas	Fall 2015-Spring 2018
<i>Co-Organizer</i> , KUMUNU 2017	University of Kansas	October 2017
<i>Co-Organizer</i> , 2017 SACNAS, Session on CA	Salt Lake City, UT	October 2017
<i>Co-Organizer</i> , AMS Spe. Ses. on CA	University of North Texas, Denton, TX	September 2017
<i>Co-Organizer</i> , JMM - MRC Session on CA	Seattle, WA	January 2016

<i>Co-Organizer</i> , AMS Spe. Ses. on CA	Loyola University, Chicago, IL	October 2015
<i>Co-Organizer</i> , Student Comm. Alg. Seminar	Purdue University	Fall 2013-Spring 2015
<i>Co-organizer</i> , Student Colloquium	Dept. of Math., Purdue University	Fall 2013-Spring 2015
<i>Co-organizer</i> , Program Associate Seminar	MSRI	Fall 2012

OTHER PROFESSIONAL ACTIVITIES

Scientific Committee member, Colombian Combinatorics Meeting (ECCO), summer 2024
Teaching Assistant, MSRI Summer Graduate School, University of Notre Dame, May 2019
Teaching Assistant, MSRI Summer Graduate School, Okinawa, Japan, May 2017
Panelist, NSF
Reviewer, NSA-AMS Grant
Referee, Advances in Mathematics, Algebra & Number Theory, Archiv der Mathematik, Boletín de la Sociedad Matemática Mexicana, Collectanea Mathematica, Communications in Algebra, Compositio Mathematica, Contemporary Mathematics, European Journal of Combinatorics, Journal of Algebra, Journal of Algebra and its Applications, Journal of Commutative Algebra, Journal of the London Mathematical Society, Journal of Pure and Applied Algebra, Mathematical Research Letters, Mathematica Scandinavica, Mathematische Nachrichten, Mathematische Zeitschrift, Nagoya Mathematical Journal, Proceedings of the American Mathematical Society, Rocky Mountain Journal of Mathematics, Results in Mathematics, Transactions of the American Mathematical Society.

ADVISING

Postdocs:
Sandra Rodriguez-Villalobos (2026 -)
Sankhaneel Bisui (2023 - 2025) - Currently an Assistant Professor at IISER Bhopal
Ph.D. students:
J. Fletcher (2025 -)
Sudipta Das (2020 - 2025) - Currently a Postdoc at Tata Institute of Fundamental Research
Master's Degree students:
Emmy Lewis (2019 - 2021) - Currently a PhD Student at Cornell University
Undergraduate students:
Ethan Leventhal (2022 - 2024) - Currently a PhD Student at Georgia Tech
Jacob Cooper (2023 - 2024) - Currently a PhD Student at ASU
Emmy Lewis (2018 - 2019) - Currently a PhD Student at Cornell University

SELECTED OUTREACH ACTIVITIES

<i>Representative for School of Math&Stats</i> , ASU	SACNAS, Nat. Conference	2022, 2023, 2024, 2025
<i>Associate Editor</i> , AMS Blog on Teaching and Learning		August 2020 - July 2021
<i>Associate Editor</i> , Frontiers for Young Minds		May 2017 - May 2021
<i>Grader</i> , USAMO 2015, 2016	Cambridge, MA	April 2015, 2016
<i>Coordinator (Problem Captain)</i> , IMO 2013	Santa Marta, Colombia	July 2013
<i>Trainer</i> , AAAS Olympiad Program	Washington D.C.	2012
<i>President</i> , Colombian St. Association at Purdue	Purdue University	2011

SELECTED AWARDS IN MATHEMATICS AND PROGRAMMING COMPETITIONS

International Collegiate Programming Contest, World Finals, *top 50*, Sweden, 2009.
International Collegiate Programming Contest, South American/North Finals, *first place*, Bogotá, Colombia, 2008.
Colombian Programming Contest (college level), *first place*, 2008.
International Mathematics Competition for University Students, *third prize*, Bulgaria, 2008; Ukraine, 2006.
International Mathematical Olympiad, *bronze medal*, Greece, 2004.
Colombian Mathematical Olympiad, *gold medal*, 2003.
Colombian Programming Olympiad, *first place*, 2002, 2001, *third place*, 2003.

TEACHING EXPERIENCE

Arizona State University, August 2022 -

Abstract Algebra I	Fall 2026
Intermediate Abstract Algebra	Spring 2026
Ring Theory	Spring 2026
Abstract Algebra III (Alg. Geom. and Toric Var.)	Fall 2025
Abstract Algebra II	Spring 2025
Abstract Algebra I	Fall 2024
Algebraic Geometry and Toric Varieties	Fall 2023
Advanced Linear Algebra	Fall 2023
Ring Theory	Spring 2023
Advanced Linear Algebra	Fall 2022

New Mexico State University, August 2018 - July 2022

Abstract Algebra I: Groups and Rings	Spring 2022
Calculus and Analytic Geometry II	Spring 2022
Introduction to Modern Algebra	Fall 2021
Calculus and Analytic Geometry II	Fall 2021
Introduction to Algebraic Statistics	Spring 2021
Introduction to Higher Mathematics	Spring 2021
Advanced Linear Algebra	Fall 2020
Calculus and Analytic Geometry II	Fall 2020
Introduction to Algebraic Geometry	Spring 2020
Matrix Theory and Applied Linear Algebra	Fall 2019
Calculus and Analytic Geometry III	Fall 2019
Calculus and Analytic Geometry II	Spring 2019
Calculus and Analytic Geometry I (two sections)	Fall 2018

University of Kansas, August 2015 - May 2018

Calculus III (Honors)	Spring 2018
Introductory Modern Algebra	Fall 2017
Elementary Linear Algebra	Fall 2017
Linear Algebra	Summer 2017
Linear Algebra	Spring 2017
Elementary Linear Algebra	Fall 2016
Calculus III (Honors)	Fall 2016
Elementary Linear Algebra	Spring 2016
Vector Calculus	Spring 2016
Linear Algebra	Fall 2015

Purdue University, August 2009 - July 2015

Introductory Analysis I	Spring 2014
Introductory Analysis II	Fall 2013
Calculus I, recitation	Spring 2013
Calculus II, recitation	Spring 2012
Multivariable Calculus, recitation	Fall 2011
Multivariable Calculus, recitation	Fall 2010

Universidad Nacional de Colombia, Bogotá, August 2008 - August 2009

Linear Algebra	Spring 2009
Calculus I	Fall 2008

LANGUAGES

English (bilingual proficiency), Spanish (native).